

5_minute_buckets

January 7, 2025

```
[1]: %%capture
    ## general setup
    %load_ext sql
    %sql duckdb://
    %config SqlMagic.named_parameters="enabled"
    %config SqlMagic.displaylimit = None
    import sys
    sys.path.append(".")

    ## args
    source = '/tmp/warehouse_oa/*.parquet' # where the query starts

    ## sql snippets
    ## each snippet create a virtual table
    # import snippets
    from snippets.base_events import get_base_events_snippet
    from snippets.zone_pairs import get_zone_pairs_snippet
    from snippets.visit_durations import get_visit_durations_snippet
    from snippets.aggregated_data import get_aggregated_data_snippet
    from snippets.merged_data import get_merged_data_snippet

    # generate snippets
    base_events_snippet = get_base_events_snippet(source)
    zone_pairs_snippet = get_zone_pairs_snippet()
    visit_durations_snippet = get_visit_durations_snippet()
    aggregated_data_snippet = get_aggregated_data_snippet()

    # save snippets in duckdb
    %sql {{base_events_snippet}} --save base_events --no-execute
    %sql {{zone_pairs_snippet}} --with base_events --save zone_pairs --no-execute
    %sql {{visit_durations_snippet}} --with zone_pairs --save visit_durations
    ↪--no-execute
    %sql {{aggregated_data_snippet}} --with visit_durations --save aggregated_data
    ↪--no-execute

    _zones = %sql --with visit_durations SELECT DISTINCT zone_name FROM
    ↪visit_durations
```

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zones = [zone[0] for zone in _zones]

merged_data_snippet = get_merged_data_snippet(zones)
%sql {{merged_data_snippet}} --with aggregated_data --save merged_data_
↪--no-execute

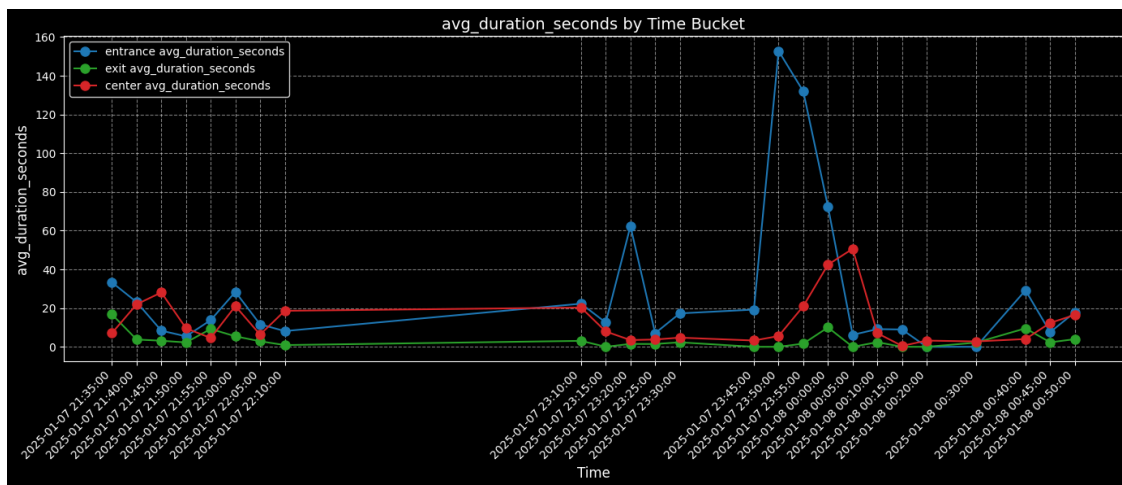
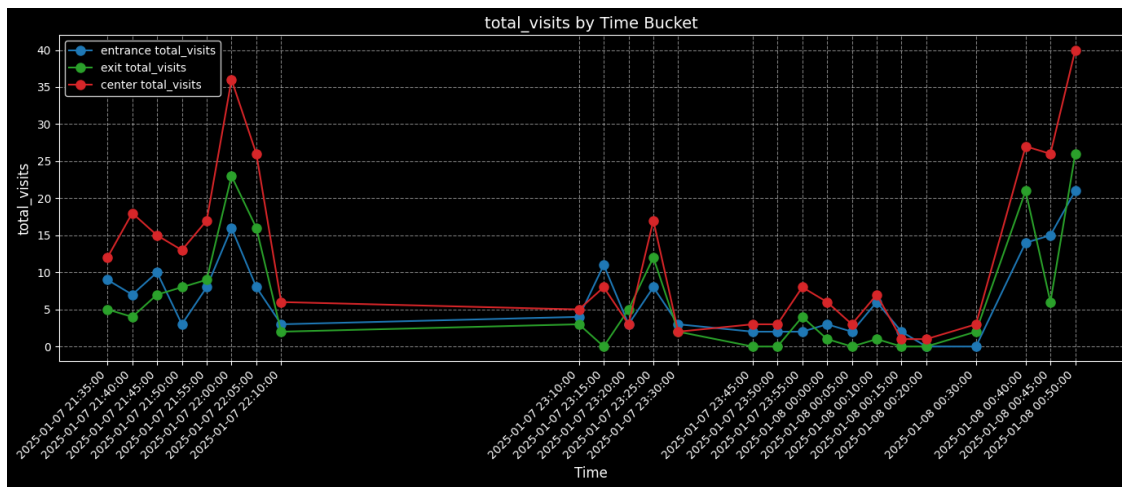
## get the data
data = %sql --with merged_data select * from merged_data LIMIT 100000

```

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[2]: ## plot
from plot import plot
plot(data, zones, 'total_visits')
plot(data, zones, 'avg_duration_seconds')

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[3]: %%sql --with merged_data
select * from merged_data LIMIT 100000
```

Running query in 'duckdb://'

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[3]: +-----+-----+-----+-----+
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|      time_bucket      | entrance_total_visits | entrance_avg_duration_seconds |
exit_total_visits | exit_avg_duration_seconds | center_total_visits |
center_avg_duration_seconds |
+-----+-----+-----+-----+
-----+
| 2025-01-07 21:35:00 |          9          |          33.27          |
5          |          16.73          |          12          |          7.09
|
| 2025-01-07 21:40:00 |          7          |          23.08          |
4          |          3.77          |          18          |          21.95
|
| 2025-01-07 21:45:00 |         10         |          8.33          |
7          |          3.14          |          15          |          28.08
|
| 2025-01-07 21:50:00 |          3          |          5.42          |
8          |          2.25          |          13          |          9.57
|
| 2025-01-07 21:55:00 |          8          |         13.78          |
9          |          9.21          |          17          |          4.55
|
| 2025-01-07 22:00:00 |         16         |         28.37          |
23         |          5.42          |          36          |          21.03
|
| 2025-01-07 22:05:00 |          8          |         11.61          |
16         |          2.86          |          26          |          6.3
|
| 2025-01-07 22:10:00 |          3          |          8.13          |
2          |          0.9          |          6          |          18.6
|
| 2025-01-07 23:10:00 |          4          |         22.28          |
3          |          3.08          |          5          |          20.27
|
| 2025-01-07 23:15:00 |         11         |         12.54          |
0          |          0.0          |          8          |          7.92
|
| 2025-01-07 23:20:00 |          3          |         62.35          |
5          |          1.5          |          3          |          3.48
|
| 2025-01-07 23:25:00 |          8          |          7.01          |
```

12		1.46		17		3.77
	2025-01-07 23:30:00		3		17.27	
2		2.32		2		4.74
	2025-01-07 23:45:00		2		19.21	
0		0.0		3		3.25
	2025-01-07 23:50:00		2		152.81	
0		0.0		3		5.55
	2025-01-07 23:55:00		2		131.95	
4		1.63		8		20.94
	2025-01-08 00:00:00		3		72.41	
1		10.01		6		42.41
	2025-01-08 00:05:00		2		6.18	
0		0.0		3		50.61
	2025-01-08 00:10:00		6		9.15	
1		2.4		7		7.03
	2025-01-08 00:15:00		2		8.94	
0		0.0		1		0.48
	2025-01-08 00:20:00		0		0.0	
0		0.0		1		3.21
	2025-01-08 00:30:00		0		0.0	
2		2.16		3		2.78
	2025-01-08 00:40:00		14		29.2	
21		9.54		27		4.01
	2025-01-08 00:45:00		15		7.56	
6		2.32		26		12.35
	2025-01-08 00:50:00		21		17.99	
28		3.71		40		16.57
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